

Lecture 9: Limits and Continuity in Several Variables

Background & Motivation

Before we can differentiate multivariate functions, we need limits. The key new difficulty: in one variable, you approach a point from the left or right. In two variables, you can approach from infinitely many directions. This makes proving limits exist much harder—and gives us a powerful tool (path analysis) for showing they don't.

1. Multivariable Limits

Definition 1: Limit of $f(x, y)$ as $(x, y) \rightarrow (a, b)$

We write $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$ if $f(x, y)$ can be made arbitrarily close to L for all (x, y) sufficiently close to (a, b) .

Key difference from single-variable:

- In 1D – only **two directions** (left and right)
- In 2D – must approach from **infinitely many directions and paths**
- If *any two paths* give different limits $\implies$ limit **DNE**

Properties of Multivariable Limits

If $\lim_{(x,y) \rightarrow (a,b)} f = L$ and $\lim_{(x,y) \rightarrow (a,b)} g = M$ exist:

- $\lim(f \pm g) = L \pm M$
- $\lim(cf) = cL$
- $\lim(fg) = LM$
- $\lim(f/g) = L/M$ if $M \neq 0$

Polynomials and rational functions (where defined) – use **direct substitution**.

2. Techniques for Evaluating Limits

Technique 1: Direct Substitution

– Always try first; works when f is continuous at (a, b)

Technique 2: Algebraic Simplification / Rationalization

– Factor, multiply by conjugate, or simplify before substituting

Example 1: Rationalization

Evaluate $\lim_{(x,y) \rightarrow (1,1)} \frac{\sqrt{x} - \sqrt{y}}{x - y}$.

1. Rationalize:

$$\frac{\sqrt{x} - \sqrt{y}}{x - y} \cdot \frac{\sqrt{x} + \sqrt{y}}{\sqrt{x} + \sqrt{y}} = \frac{x - y}{(x - y)(\sqrt{x} + \sqrt{y})} = \frac{1}{\sqrt{x} + \sqrt{y}}$$

2. Substitute:

$$\lim_{(x,y) \rightarrow (1,1)} \frac{1}{\sqrt{x} + \sqrt{y}} = \frac{1}{1 + 1} = \boxed{\frac{1}{2}}$$

Technique 3: Polar / Spherical Coordinates

- For limits at the origin: $x = r \cos \theta$, $y = r \sin \theta$, $(x, y) \rightarrow (0, 0) \iff r \rightarrow 0^+$
- If the result is independent of θ and tends to L as $r \rightarrow 0$, the limit is L

Example 2: Polar Coordinates

Evaluate $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{\sin(x^2 + y^2 + z^2)}{x^2 + y^2 + z^2}$.

1. Convert: In spherical coordinates, $x^2 + y^2 + z^2 = \rho^2$

2. Simplify: $\lim_{\rho \rightarrow 0^+} \frac{\sin(\rho^2)}{\rho^2}$

3. Evaluate: Let $u = \rho^2 \rightarrow 0$: $\lim_{u \rightarrow 0} \frac{\sin u}{u} = \boxed{1}$

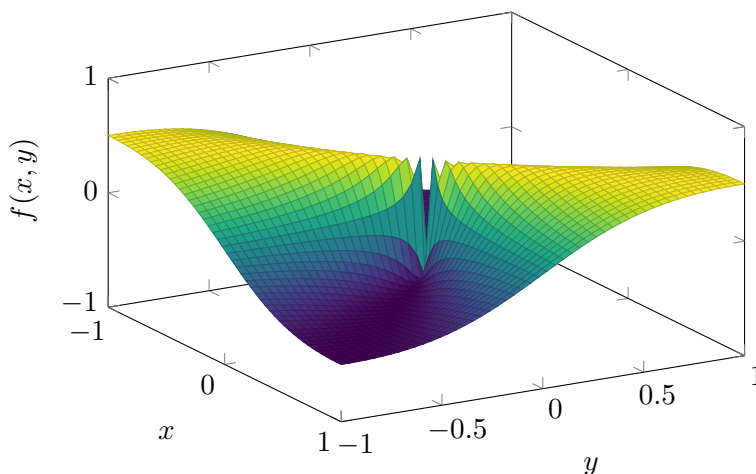
Technique 4: Path Analysis (for showing DNE)

- If two different paths give different limiting values $\implies$ limit DNE

Common Paths to Try

- Along axes: $y = 0$, then $x = 0$
- Along lines: $y = mx$ for various m
- Along curves: $y = x^2$, $y = x^3$, etc.

If all linear paths give the same value – try a curved path before concluding the limit exists!



Graph of $z = \frac{xy}{x^2 + y^2}$ near the origin. The surface takes different limiting values along different paths to $(0, 0)$: along $y = 0$ or $x = 0$ it equals 0, but along $y = x$ it equals $1/2$. Hence the limit does not exist.

Example 3: Limit DNE by Path Analysis

Show $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.

1. **Path along $y = 0$:** $\frac{x^2 - 0}{x^2 + 0} = 1 \implies \text{limit} = 1$
2. **Path along $x = 0$:** $\frac{0 - y^2}{0 + y^2} = -1 \implies \text{limit} = -1$
3. **Result:** $1 \neq -1$, so the limit **DNE**.

Example 4: Tricky: All Lines Give 0, But Limit DNE

Does $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}$ exist?

1. **Along $y = mx$:** $\frac{x \cdot m^2 x^2}{x^2 + m^4 x^4} = \frac{m^2 x}{1 + m^4 x^2} \rightarrow 0$ as $x \rightarrow 0$
Every straight line gives limit = 0.
2. **Along $x = y^2$:** $\frac{y^2 \cdot y^2}{y^4 + y^4} = \frac{y^4}{2y^4} = \frac{1}{2}$
3. **Result:** $0 \neq 1/2$, so the limit **DNE**.

Lesson: Checking only straight-line paths is not sufficient!

3. Continuity

Definition 2: Continuity of $f(x, y)$

f is **continuous at (a, b)** if all three hold:

1. $f(a, b)$ is defined
2. $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ exists
3. $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$

Polynomials, exponentials, trig functions, and their compositions are continuous on their domains.

Practice Problems

Problem 1. Evaluate $\lim_{(x,y) \rightarrow (2,3)} (x^2 y + xy^2 - 1)$.

Answer: $(4)(3) + (2)(9) - 1 = 12 + 18 - 1 = 29$

Problem 2. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$.

Polar: $\frac{r^2 \cos^2 \theta \cdot r \sin \theta}{r^2} = r \cos^2 \theta \sin \theta$

$|r \cos^2 \theta \sin \theta| \leq r \rightarrow 0$, so the limit is $\boxed{0}$ by Squeeze Theorem.

Problem 3. Show $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$ does not exist.

Along $y = 0$: $0/x^2 = 0$, limit = 0.

Along $y = x$: $x^2/(2x^2) = 1/2$, limit = $1/2$.

$0 \neq 1/2 \implies$ DNE.

Problem 4. Evaluate $\lim_{(x,y) \rightarrow (1,1)} \frac{x^2 - y^2}{x - y}$.

Factor: $\frac{(x-y)(x+y)}{x-y} = x + y \rightarrow 1 + 1 = \boxed{2}$ (cancel since $x \neq y$ in the limit).

Problem 5. Show $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 y^2 + (x - y)^2}$ does not exist.

Along $y = 0$: $\frac{0}{0+x^2} = 0$, limit = 0.

Along $y = x$: $\frac{x^4}{x^4+0} = 1$, limit = 1.

$0 \neq 1 \implies$ DNE.

Problem 6. Prove using ϵ - δ that $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2 y}{x^2 + y^2} = 0$.

Hint: Use $|y| \leq \sqrt{x^2 + y^2}$.

$$\left| \frac{3x^2 y}{x^2 + y^2} \right| = 3 \cdot \frac{x^2}{x^2 + y^2} \cdot |y| \leq 3 \cdot 1 \cdot |y| \leq 3\sqrt{x^2 + y^2}$$

Given $\epsilon > 0$, choose $\delta = \epsilon/3$. If $\sqrt{x^2 + y^2} < \delta$:

$$\left| \frac{3x^2 y}{x^2 + y^2} \right| \leq 3\sqrt{x^2 + y^2} < 3\delta = \epsilon. \quad \square$$